

Harsh Bajaj

Backend Developer | AI-Powered Systems, Scalable APIs & Semantic Search | Node.js, Redis, NLP, System Design

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TECHNICAL SKILLS

Backend: Node.js, Express.js, Python, REST APIs, Redis, BullMQ, PostgreSQL, MongoDB

AI / NLP: Semantic Search, TF-IDF, Vector Embeddings, ONNX Runtime, Cosine Similarity

Infrastructure: Docker, Linux, Nginx, PM2, Load Testing, Async Processing

Security & Tools: JWT, RBAC, Rate Limiting, Git, Postman, Puppeteer

EXPERIENCE

Web Developer

Apr 2025 – Apr 2026

Cynox Security LLP

Delhi, India

- Architected and delivered a production-grade Task Management System by converting loosely defined stakeholder requirements into scalable backend services and APIs.
- Improved system throughput by $\sim 2\times$ through load testing, optimized database pooling, and asynchronous processing using Redis and BullMQ.
- Stabilized VPS infrastructure by diagnosing malicious resource consumption and redesigned PDF generation workflows to reduce report size by $\sim 10\times$.

Software Engineer

Nov 2023 – Sep 2024

North Delhi Public School

Delhi, India

- Processed and structured institutional datasets to support reporting workflows and operational insights.
- Developed backend-driven workflows using PHP and WordPress for enquiry management and data collection systems.

PROJECTS

AI Resume Matching & ATS Feedback System

GitHub

- Designed a tiered semantic matching engine using ONNX Runtime embeddings (all-MiniLM-L6-v2) with TF-IDF fallback pipelines for resilient inference and contextual resume-job alignment.
- Built an inflation-resistant scoring architecture combining semantic similarity, domain alignment weighting, and penalty engines to reduce keyword stuffing and inaccurate candidate scoring.
- Optimized offline TF-IDF training on large resume-job corpora (50,000+ n-gram features) and load tested the platform at 600+ req/sec with sub-300ms p95 latency.

Task Management System

Live | GitHub

- Built a production-grade task management platform with modular backend architecture, JWT authentication, RBAC, and structured middleware pipelines.
- Improved responsiveness and throughput using Redis and BullMQ for asynchronous background processing and optimized database query handling.
- Implemented rate limiting, request validation, and traffic handling controls to improve API stability under concurrent usage.

EDUCATION

Indira Gandhi National Open University (IGNOU)

2025 – 2027

Master of Computer Applications (MCA)

New Delhi

Deshbandhu College

2021 – 2025

BSc Computer Science

Delhi